

Splitting the 12V line is only half the battle; you must also manage the ground return paths.

When those three motors pull 6A, that 6A has to flow back to the battery through the ground plane. If that 6A flows underneath or through the ground copper near your STM32 or your TPS56637 feedback resistors, it will raise the local ground voltage, causing your microcontroller to read false button presses, corrupt SPI data, or trigger a brown-out reset.

How to route it in Altium:

Do not use one massive, uniform ground pour for the whole board.

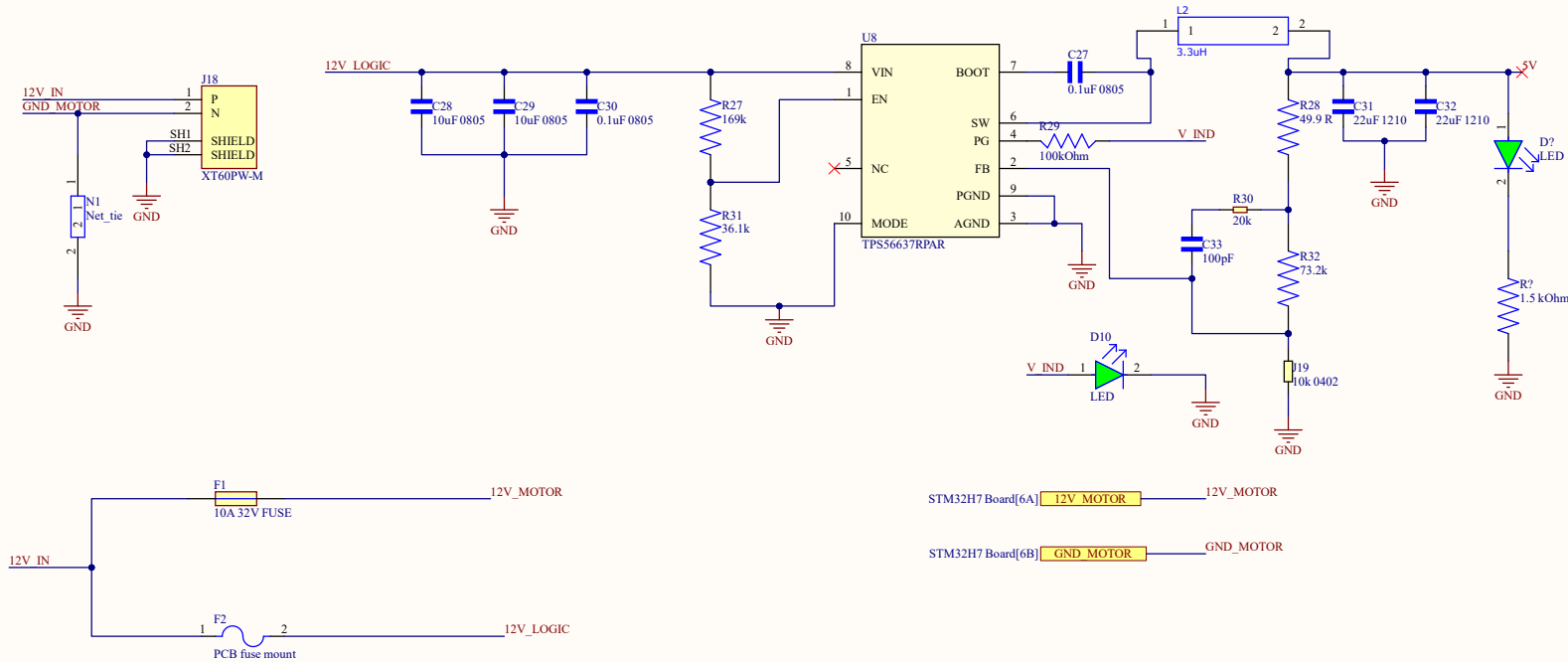
Create a specific, thick ground trace (or dedicated polygon pour) that connects the GND pins of the three TMC2209s directly back to the PGND pin of the XT60 connector.

Keep this motor ground return path physically away from your STM32 and analog sensors.

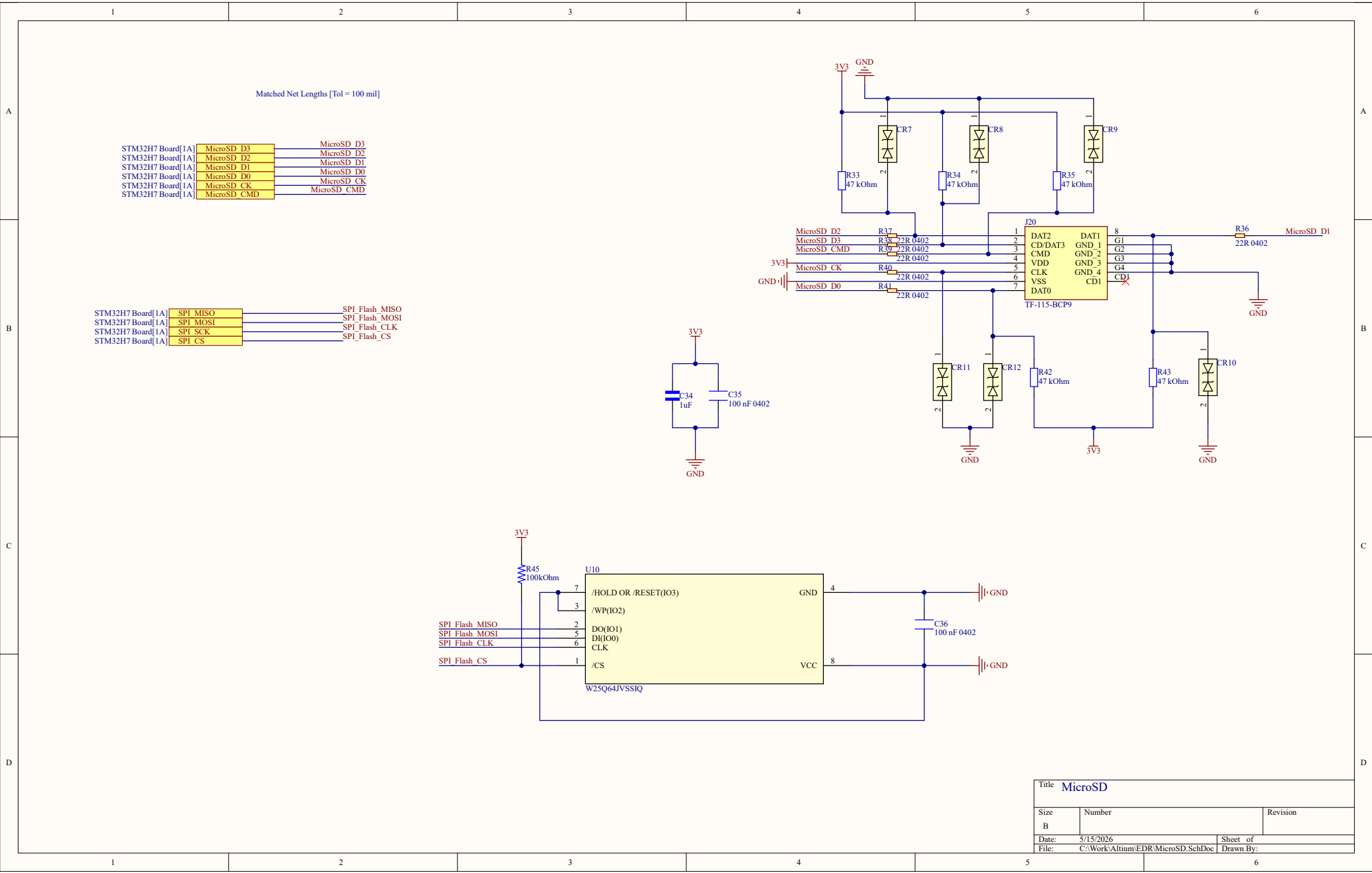
Your logic ground (PGND and AGND from the buck converter, and the STM32 ground) should meet the motor ground at exactly one single point: the physical solder pad of the XT60's negative pin.

3. Don't Forget the Bulk Capacitors!

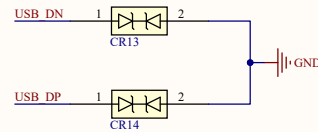
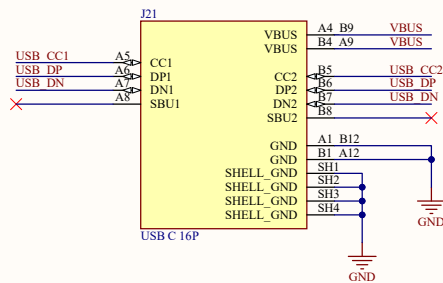
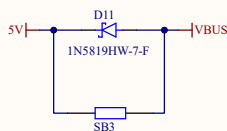
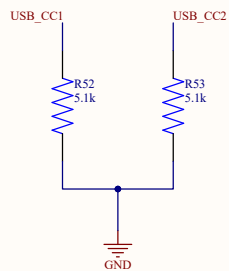
As a final reminder, because you are running 3 motors from this 12V line, you absolutely must place a large aluminum electrolytic capacitor (e.g., 220µF to 470µF, rated for 35V or 50V) on the 12V MOTORS net right next to the TMC2209 drivers. Without it, the inductive flyback from three simultaneous 2A motors will easily spike the 12V rail past 30V and detonate your chips.



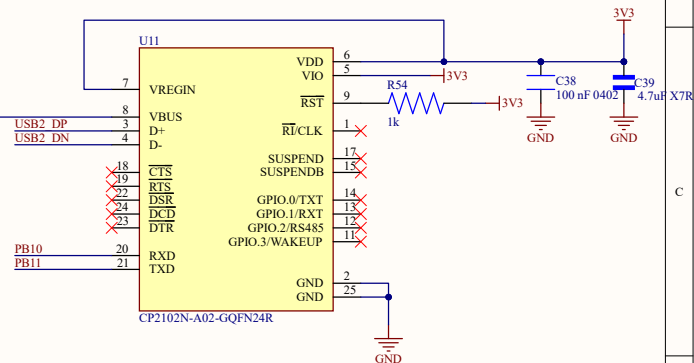
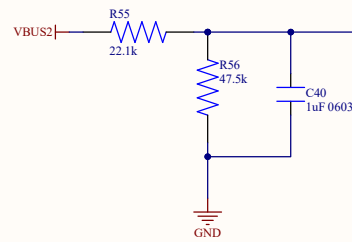
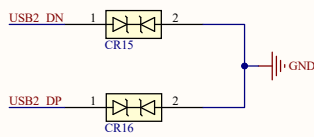
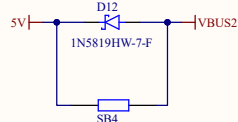
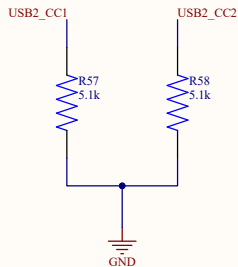
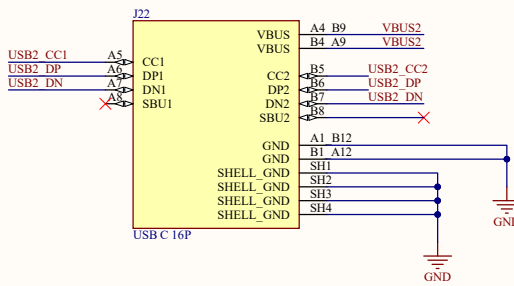
Title: POWER		
Title B	Number	Revision
Date: File:	5/15/2026 C:\Work\Altium\EDR\POWER.SchDoc	Sheet of Drawn By:



STM32H7 Board[1A] USB1_DN USB_DN
STM32H7 Board[1A] USB1_DP USB_DP



STM32H7 Board[1A] USB2_TX PB10
STM32H7 Board[1A] USB2_RX PB11



Title		
Size	Number	Revision
B		
Date:	5/15/2026	Sheet of
File:	C:\Work\Atrium\EDR\PI CON.SchDoc	Drawn By: